



Relationship between the state-space and transfer function model:

In the state-space formulation, a dynamical system is defined in the following way

$$\left. \begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned} \right\} (1)$$

Let's take Laplace transform of equation (1).

$$\left. \begin{aligned} sX(s) - x(0) &= AX(s) + BU(s) \\ Y(s) &= CX(s) + DU(s) \end{aligned} \right\}$$

with zero initial condition $x(0)=0$

As a result,

$$sX(s) = AX(s) + BU(s)$$

$$\Rightarrow sX(s) - AX(s) = BU(s)$$

$$\Rightarrow [sI - A]X(s) = BU(s)$$

$$\Rightarrow X(s) = [sI - A]^{-1}BU(s)$$

Insert the value of $X(s)$ into
the second equation

$$\rightarrow Y(s) = C[sI - A]^{-1}BU(s) + DU(s)$$

$$\Rightarrow \frac{Y(s)}{U(s)} = C[sI - A]^{-1}B + D$$

$$\Rightarrow G(s) = C[sI - A]^{-1}B + D$$

$$\Rightarrow G(s) = C \cdot \frac{1}{[sI - A]} \cdot B + D$$

In order for the transfer function $G(s) \rightarrow \infty$, the determinant

$$\det[sI - A] = 0.$$

We will now observe that the determinant of $[sI - A]$ provides the eigenvalues of matrix A .

The eigenvalue problem can be described by the following equation:

$$Ax = \lambda x$$

This equation refers to the fact that there are some "eigen" or "special" vector x , which when multiplied by the matrix A , produces the same "special" vector x . Additionally, there is a scaling factor λ . λ is called the "eigenvalues" and x is called the "eigenvector".

In order to find the eigen value and eigen vector x , let's proceed

in the following way:

$$Ax = \lambda x$$

$$\Rightarrow Ax - \lambda x = 0$$

$$\Rightarrow \lambda x - Ax = 0$$

$$\Rightarrow \lambda I x - Ax = 0$$

$$\Rightarrow (\lambda I - A)x = 0$$

$$\text{If } x \neq 0, \boxed{\det(\lambda I - A) = 0}$$

This is also called the characteristic equation. Compare this equation with the $\boxed{\det(sI - A) = 0}$ in the state space equation.

As a result, the eigenvalues of the matrix A are the poles of the transfer function.

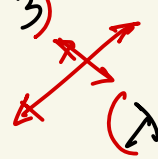
As an example, let's find the eigenvalues and eigenvectors of a matrix. Let's assume that, you happen to derive the system matrix A as follows:

$$A = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

$$\Rightarrow \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} (\lambda-3) & 1 \\ 1 & (\lambda-3) \end{bmatrix} = 0$$


$$\Rightarrow (\lambda-3)^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 6\lambda + 9 - 1 = 0$$

$$\Rightarrow \lambda^2 - 6\lambda + 8 = 0$$

$$\Rightarrow (\lambda-2)(\lambda-4) = 0$$

$$\Rightarrow \lambda = 2, \lambda = 4$$

The eigenvalues are positive, so the system is unstable.

Let's find the corresponding eigenvectors.

$$\text{For } \lambda = 2,$$

$$(2I - A)x = 0$$

$$\Rightarrow \left(2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} \right) x = 0$$

$$\Rightarrow \left(\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} \right) x = 0$$

$$\Rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\Rightarrow -x_1 + x_2 = 0$$

$$\Rightarrow x_1 = x_2$$

As a result,

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The eigenvector corresponding to $\lambda=2$,

$$\text{is } g = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

let's verify:

$$Ax = \lambda x$$

$$\Rightarrow \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} (3)(1) + (-1)(1) \\ (-1)(1) + (3)(1) \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$= 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \lambda x$$

$\downarrow$
 $= 2$

so we found a special vector

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Similarly, for $\lambda = 4$,

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} x_1 \quad \left[\begin{array}{l} \text{By following the} \\ \text{same process as} \\ \lambda = 2 \end{array} \right]$$